KOSALA NAYANAJITH DESHAPRIYA

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PROFESSIONAL SUMMARY

Results-driven Computer Engineering undergraduate with 4+ years of freelance experience in web development and graphic design. Specializes in machine learning, deep learning, computer vision, and MLOps, with active research on CNN-based agricultural AI. Proven ability to build end-to-end ML pipelines, deploy IoT-integrated AI systems, and develop full-stack web applications. Actively seeking remote opportunities in AI/ML Engineering, Data Science, MLOps, and DevOps.

EDUCATION

B.Sc.Eng (Hons) in Computer Engineering - University of Jaffna, Sri Lanka *May 2022 – 2026*

G.C.E. A/L: Combined Maths – B, Chemistry – B, Physics – B | Z-Score: 1.5200

Relevant Coursework: Machine Learning • Intelligent Systems • Bioinformatics • Data Structures & Algorithms

RESEARCH

Tomato Leaf Density-Based Growth Detection with Deep Learning *2025 – Present*

- Designed a dual-branch deep learning architecture combining Attention U-Net for leaf segmentation and a CNN for growth-stage classification of tomato plants in greenhouse conditions.
- Developed a Leaf Pixel Fraction (LPF) metric to quantify leaf density, fused with CNN visual features to classify four growth stages: Seedling, Vegetative, Flowering, and Fruiting.
- Annotated a tomato plant image dataset using Roboflow with polygon masks in COCO JSON format; applied train/validation/test splits for model training.
- Applied transfer learning with pre-trained CNN backbones, trained using the Adam optimizer with image preprocessing at 224×224 resolution.

Tools: **Python · PyTorch · OpenCV · TensorFlow/Keras · CNN · LSTM · Transfer Learning · CI/CD**

PROJECTS

FlashGuard 1.0 - AI-Assisted Embedded Flash Flood Early Warning System *2025*

- Designed and built a real-time flood risk classification system for the Kelani River, Sri Lanka, using a Hall Effect flow sensor, ATmega328P, and ESP32.
- Deployed a TensorFlow Lite model on-device to classify water velocity into Safe, Warning, or Critical levels with sub-second latency.
- Built a cloud-connected web dashboard with push notifications (Firebase + Vercel) and automated alerting via n8n workflows.

Tools: **ESP32 · ATmega328P · TensorFlow Lite · n8n · Firebase · Vercel · IoT · Hall Effect Sensor**

Movie Recommendation System - AI-Powered Content-Based Recommender *Oct 2025*

- Built a content-based movie recommender using TF-IDF Vectorization and Cosine Similarity to surface similar titles based on genre and textual features.
- Developed an interactive Gradio web app; optimized model storage using HDF5 and Joblib for fast inference on large datasets.

Tools: Python · Pandas · NumPy · Scikit-learn · Gradio · TF-IDF · Cosine Similarity · H5py

Heart Disease Predictor - ML-Powered Healthcare Web Application *Jul – Aug 2025*

- Built a heart disease prediction web app applying multiple ML algorithms including Logistic Regression, Random Forest, and ensemble methods on clinical patient data.
- Conducted end-to-end data analysis, feature engineering, and model comparison to optimize predictive accuracy for cardiology risk classification.
- Deployed as a live web application on Heroku with a JavaScript frontend for real-time predictions.

Tools: **Python · Scikit-learn · Logistic Regression · Random Forest · NumPy · Flask · JavaScript · Heroku · Git**

YakaWork - Sri Lankan Freelancer Marketplace Platform *Ongoing*

- Architecting a full-stack freelancer marketplace connecting local talent with clients, featuring authentication, service listings, messaging, and Sri Lankan payment gateway integration.

Tools: **Next.js · React · Node.js · REST APIs**

Laptop Price Predictor - End-to-End ML Deployment *2024*

- Built a full ML pipeline (EDA → feature engineering → Random Forest Regression → evaluation) and deployed as a live web app via Flask.

Tools: **Python · Scikit-learn · Random Forest · Flask · HTML/CSS**

EncHAT - Real-Time Encrypted Chat Platform *July 2026*

- Architecting a secure communication platform with end-to-end message encryption and planned WebRTC video call integration.

Tools: **WebRTC · End-to-End Encryption**

TECHNICAL SKILLS

ML / AI / DL: PyTorch · TensorFlow · TFLite · Scikit-learn · OpenCV · CNN · LSTM · Transfer Learning
MLOps / DevOps: Git · GitHub Actions · CI/CD · Docker · Vercel · Firebase · Linux · REST APIs · Model Versioning
Data Science: Python · Pandas · NumPy · Matplotlib · Feature Engineering · Random Forest · Regression
IoT / Embedded: ESP32 · ATmega328P · Hall Effect Sensors · TensorFlow Lite · n8n Automation
Web & Frameworks: React · Next.js · Flask · Node.js · Bootstrap · WordPress · WebRTC
Languages: Python · C · Java · TypeScript · JavaScript · PHP · HTML5 · CSS3
Design & Tools: Git · GitHub · Docker · Postman · ROS · Jira

CERTIFICATIONS

Certification	Issuer	Date
Machine Learning with Python	IBM (Coursera)	Jun 2026
Introduction to Artificial Intelligence (AI)	IBM (Coursera)	Jun 2026
AWS S3 Basics	Coursera	May 2026
UOJ Coders Version 4.0	CompSoc-UOJ	Aug 2025
BITCODE V5.0 – Inter University Coding Competition	Rajarata Univ. of Sri Lanka	Nov 2024
HTML, CSS & JavaScript for Web Developers	Johns Hopkins Univ. (Coursera)	Oct 2024

CONTENT CREATION

YouTube Channel - Kosala Nayanajith Deshapriya youtube.com/@kosaladeshapriya4027

Publishes tutorials and insights on machine learning and emerging technology trends for a growing audience.

ADDITIONAL INFORMATION

Soft Skills

Project Management • Time Management • Teamwork • Fast Learner • Communication • Problem Solving • Content Creation

Languages

Sinhala – Native
English – Professional Working Proficiency
Tamil – Limited Working Proficiency

REFERENCES

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